
ASG AIRLINES

End-to-End Data Engineering Pipeline

Technical Documentation & Solution Walkthrough

Data Engineering Internship Assignment

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1. Executive Summary

This technical report presents an end-to-end data engineering pipeline developed for ASG Airlines. The primary objective of the project is to ingest four operational datasets—Flights, Bookings, Payments, and Passengers—from an enterprise Excel source, execute comprehensive data profiling, apply rigorous cleaning and standardization, perform business transformations (such as airline prefix derivation and cross-midnight flight duration validation), implement robust Personally Identifiable Information (PII) hashing and masking, and produce an aggregation-ready analytical dataset tailored for Power BI business intelligence reporting.

The technical architecture utilizes a local Python and Pandas processing workflow. This implementation choice strictly aligns with the internship assignment requirements as an approved, self-contained alternative to cloud infrastructure (e.g., Microsoft Azure) when local execution is specified. The resulting analytical dataset enforces operational data integrity and compliance by strictly excluding raw identity attributes from the BI consumption layer while enabling robust multi-dimensional analysis.

Key Project Highlights:

- **Data Source:** Excel Workbook containing 4 core relational tables (1,020 Flights, 1,000 Bookings, 1,000 Payments, 1,039 Passengers).
- **Core Technology:** Python 3.x, Pandas, NumPy, Microsoft Power BI.
- **Governance & Privacy:** SHA-256 hashing for passenger keys; masking for Aadhaar, phone, and email fields.
- **Analytical Layer:** 3-page interactive Power BI dashboard (Executive Overview, Duration & Anomalies, Route Performance).

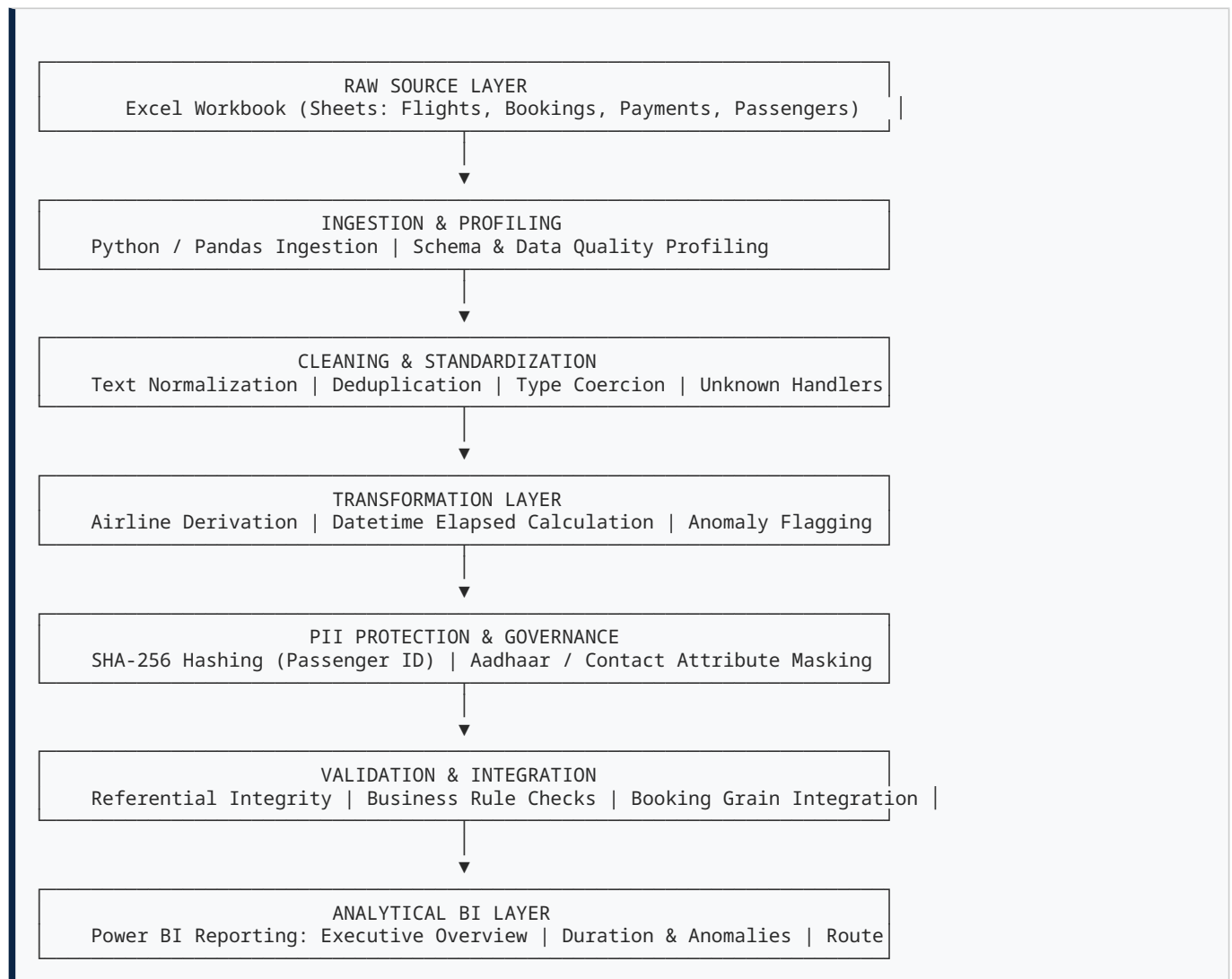
2. Objectives and Requirements

The technical implementation was designed to satisfy explicit functional and architectural requirements demanded by enterprise data engineering workflows:

- **Pipeline Architecture:** Build a complete, automated workflow spanning ingestion, data profiling, cleaning, transformation, validation, PII protection, and analytical output generation.
- **Data Quality Remediation:** Identify and systematically resolve missing values, exact duplicate records, malformed identifiers, non-numeric financial amounts, and schema inconsistencies across all operational source tables.
- **Flight Duration & Overnight Analytics:** Calculate precise elapsed flight durations from departure and arrival timestamps, handling cross-day/overnight flights seamlessly using complete datetime logic.
- **Fallback Airline Derivation:** Extract 2-character IATA-style airline prefixes from flight identifiers to resolve missing or 'UNKNOWN' source airline attributes.
- **Privacy & Governance Compliance:** Apply strict PII protection protocols (SHA-256 hashing and character masking) to guarantee that non-essential personal identifiers never reach the reporting layer.
- **Analytical Modeling & BI Enablement:** Integrate disparate operational entities into a unified, booking-grained data model structured for seamless consumption in Power BI.
- **Business KPI Support:** Accurately compute core metrics including Average Flight Duration, Route-wise Traffic, Duration Anomalies, and Airline Distribution.

3. Solution Architecture

The solution adheres to a multi-layered ETL (Extract, Transform, Load) architectural pattern. Raw operational data is preserved in an immutable staging layer, while downstream transformations progressively refine, validate, and secure the data before loading into the analytical layer.



Scalability Note: While this project is executed locally using Python and Pandas, the pipeline stages are designed with pure functional boundaries. In an enterprise cloud environment, these exact logical steps can be seamlessly migrated to PySpark on Databricks, Azure Data Factory, or Microsoft Fabric for multi-terabyte processing.

4. Data Flow Diagram

The data flow illustrates how individual operational datasets move independently through extraction, cleaning, and transformation, before converging into the integrated analytical model.

Source Dataset	Transformation & Control Processing	Target Output Entity
Flights	Profile → Remove 15 Exact Duplicates → Standardize Text → Derive Airline Prefix → Calculate Elapsed Timestamp Duration → Anomaly Threshold Flagging	Clean Flight Dimension

Source Dataset	Transformation & Control Processing	Target Output Entity
Bookings	Normalize IDs & Status → Handle Missing Status as 'Unknown' → Validate Foreign Keys (Passenger & Flight)	Protected Booking Fact
Payments	Strip Non-numeric Formatting → Coerce Invalid/Missing Amounts to Null → Validate Booking Reference	Clean Payment Dimension
Passengers	Standardize Text → Replace Missing Last Names with 'Unknown' → SHA-256 Hash Passenger Key → Mask Aadhaar/Contact Attributes	Protected Passenger Dimension
Integrated Layer	Merge Dimensions onto Booking Fact Grain → Run Referential Integrity Checks → Generate Aggregations	Analytical BI Dataset

5. Source Dataset Structure

The input dataset consists of an Excel workbook containing four distinct operational sheets. Data profiling was conducted immediately upon ingestion to establish initial volume and structural baselines.

Dataset Name	Row Count	Primary Fields	Business Purpose / Description
Flights	1,020	flight_id, airline, source, destination, departure_time, arrival_time, duration	Contains flight operational schedules, routes, and timing.
Bookings	1,000	booking_id, passenger_id, flight_id, booking_date, status, travel_details	Captures customer reservation records and booking statuses.
Payments	1,000	payment_id, booking_id, amount, payment_method	Records transaction financial amounts and payment channels.
Passengers	1,039	passenger_id, name, age, gender, contact/identity attributes	Stores passenger demographic, contact, and government identity information.

6. Data Quality Assessment

Initial data profiling revealed several key data quality issues across all four sheets. These empirical statistics established the exact requirements for the cleaning and transformation stages.

Dataset	Target Column	Observed Quality Issue	Applied Treatment Strategy
Flights	Entire Row	15 exact duplicate rows present in raw sheet	Deduplicate exact duplicate records, retaining single instance.
Flights	airline	41 null/missing values; 31 'UNKNOWN' string values	Derive airline name from standardized flight ID 2-character prefix.
Payments	amount	48 missing values; source column stored as object/string	Clean text, convert to numeric float; set non-numeric to null.
Bookings	status	45 missing/null status values	Normalize status; classify missing/invalid values as 'Unknown'.

Dataset	Target Column	Observed Quality Issue	Applied Treatment Strategy
Passengers	last_name	10 missing last name values	Preserve record integrity; assign 'Unknown' to missing last name.
Flights	duration	Source duration values inconsistent with departure/arrival	Calculate exact elapsed minutes from timestamps; compare with source.
All Sheets	Foreign Keys	Potential referential integrity breaks across tables	Execute cross-table relational validation checks before integration.

7. Data Cleaning Strategy

The cleaning strategy establishes a deterministic pipeline that ensures data consistency while preserving maximum operational information. Rather than silently dropping records with missing attributes, missing categorical fields are explicitly populated with controlled placeholders ('Unknown') to maintain analytical completeness.

- **Column Name & Text Normalization:** All header strings are converted to lowercase snake_case. Text fields undergo whitespace stripping and uppercase standardization for operational keys.
- **Exact Record Deduplication:** The 15 exact duplicate rows identified in the Flights sheet are purged, reducing total flight rows from 1,020 to 1,005 unique records.
- **Flight Identifier Standardization:** Flight IDs are trimmed and capitalized to ensure uniform formatting (e.g., 'sj-101' -> 'SJ101').
- **Financial Amount Coercion:** Payment amounts are stripped of currency symbols, converted to numeric float types, and invalid entries are coerced to null without fabricating financial values.
- **Booking Status Normalization:** Booking statuses are standardized to standard categorical states: CONFIRMED, CANCELLED, PENDING, or Unknown.
- **Passenger Name Preservation:** Passengers with missing last names are retained, populating the missing attribute with 'Unknown' to preserve complete booking history.
- **Datetime Parsing:** Timestamps across all tables are parsed into strict ISO datetime objects, coercing unparseable values to null for validation error trapping.

8. Transformation Logic

Transformation rules encode critical business logic into the data pipeline, enriching the dataset and enabling operational monitoring.

8.1 Airline Derivation

When the source airline attribute is missing (41 records) or populated as 'UNKNOWN' (31 records), the pipeline derives the true carrier identity using the first two characters of the standardized Flight ID. The authoritative mapping table applied is defined below:

Flight ID Prefix	Derived Airline Name	Source Prefix Volume (Appendix B)
SJ	SpiceJet	251 records
AI	Air India	260 records
UK	Vistara	236 records

Flight ID Prefix	Derived Airline Name	Source Prefix Volume (Appendix B)
6F	IndiGo	273 records

8.2 Flight Duration Calculation

Flight duration is calculated dynamically from validated arrival and departure timestamps using elapsed time logic:

Calculated Duration (minutes) = (Arrival Timestamp – Departure Timestamp) in Seconds / 60

Calculated durations are stored in integer minutes for computational consistency and expressed in decimal hours for executive reporting. Supplied duration values from the source are converted to minutes for validation.

8.3 Overnight Flight Handling

Flights traversing midnight present potential calculation errors if evaluated purely on time-of-day. By utilizing complete ISO datetime objects (date + time), the pipeline naturally handles cross-day flights. A flight is explicitly classified as an Overnight Flight when:

Overnight Condition: Arrival Date > Departure Date
Flight Day Type Classification: 'Overnight' if True, else 'Same Day'

8.4 Duration Anomaly Logic

To evaluate data accuracy, the pipeline compares the calculated elapsed duration against the source supplied duration. A strict 5-minute tolerance threshold is enforced:

Duration Difference (min) = | Calculated Duration – Supplied Duration |
Duration Anomaly Condition: $ABS(\text{Duration Difference}) > 5$ minutes
Classification: 'Anomaly' if True, else 'Valid'

CRITICAL TECHNICAL DISTINCTION: A 'Duration Anomaly' indicates a data quality discrepancy or recorded timing mismatch within the source workbook. It MUST NOT be confused with a 'Flight Delay'. Evaluating real-world flight delays requires comparing scheduled departure/arrival times against actual departure/arrival times—fields that do not exist in the source dataset. Maintaining this explicit distinction demonstrates technical rigor.

9. PII Protection & Access Control

To comply with modern enterprise data privacy regulations (e.g., Digital Personal Data Protection Act / GDPR), the pipeline incorporates robust PII obfuscation prior to generating the analytical BI dataset.

- **Cryptographic Hashing (SHA-256):** The primary natural key passenger_id is transformed into passenger_id_hash using the SHA-256 cryptographic hash algorithm. This provides a deterministic key for relational joins without exposing raw identifiers.
- **Government ID & Contact Masking:** Sensitive attributes are masked prior to storage:
 - **Aadhaar / Identification:** Masked to display only the last 4 digits (e.g., 'XXXX-XXXX-1234').
 - **Phone Numbers:** Masked to preserve country/area context while obscuring digits (e.g., '+91-XXXXX-67890').
 - **Email Addresses:** Masked preserving first letter and domain (e.g., 'k*****@domain.com').
- **Attribute Exclusion:** Raw identity fields, passport numbers, and emergency contact details are strictly excluded from the analytical layer exported to Power BI.

Data Governance Principle:

The reporting layer operates entirely on sanitized, hashed, and aggregated data. Raw PII remains isolated within secure staging environments accessible only by authorized data engineers.

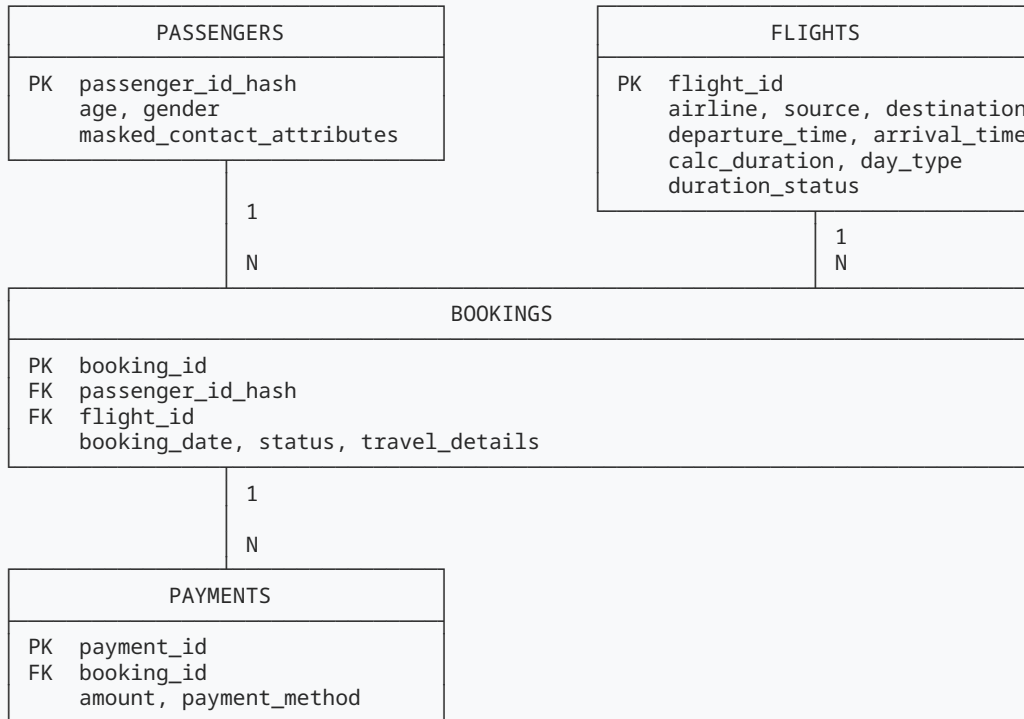
10. Data Validation & Governance

Automated validation rules execute prior to loading the data into the analytical layer. The table below details the active governance rules and their operational intent:

Validation Check	Rule / Expected Condition	Operational Governance Purpose
Flight ID Format	Matches regex <code>^(SJ AI UK 6F)\d{3,}\$</code>	Detect malformed or corrupted flight identifiers.
Airline Consistency	Derived airline equals cleaned airline	Prevent incorrect airline attribution across reports.
Timestamp Validity	Departure & Arrival parseable as ISO datetime	Ensure baseline data exists for duration logic.
Positive Duration	Calculated Duration > 0 minutes	Flag physically impossible negative or zero flight times.
Duration Anomaly	$ABS(\text{Calculated} - \text{Supplied}) \leq 5$ minutes	Isolate inconsistent source duration records.
Booking ID Uniqueness	Distinct booking_id count == Row count	Guarantee booking-level grain without duplicate explosion.
Booking -> Passenger FK	passenger_id exists in Passenger master	Enforce referential integrity for customer data.
Booking -> Flight FK	flight_id exists in Flight master	Enforce referential integrity for flight schedules.
Payment -> Booking FK	booking_id exists in Booking master	Enforce financial transaction integrity.
Payment Amount Validity	Amount is numeric and ≥ 0	Protect financial aggregation metrics from corruption.
Booking Status Validity	Status in {'CONFIRMED','CANCELLED','PENDING','Unknown'}	Standardize categorical values for BI filters.
PII Exposure Guard	Zero raw identity fields present in BI extract	Mandate compliance with data privacy policies.

11. Data Model

The analytical model is structured as a star-like relational schema centered around the Bookings entity, representing the primary operational grain of the business.



12. Business KPIs & Analytical Metrics

The integrated dataset directly powers key executive and operational metrics required by ASG Airlines management:

Metric Name	Mathematical Definition / Calculation Logic	Business & Operational Value
Average Flight Duration	AVG(Calculated Duration Minutes) grouped by Route / Airline	Monitors route efficiency and standard flight scheduling.
Route-wise Traffic	COUNT(DISTINCT booking_id) grouped by Source & Destination	Identifies high-demand travel corridors and route volume.
Duration Anomalies	COUNT(booking_id) WHERE duration_status == 'Anomaly'	Isolates data quality errors and flight timing discrepancies.
Distribution by Airline	COUNT(flight_id) and COUNT(booking_id) grouped by Airline	Evaluates market share and flight volume across carriers.
Total Bookings	COUNT(DISTINCT booking_id)	Tracks overall business reservation volume.
Total Flights	COUNT(DISTINCT flight_id)	Measures operational flight network scale.
Overnight Flights	COUNT(flight_id) WHERE flight_day_type == 'Overnight'	Monitors night-time operational flight allocation.
Total Payments	SUM(amount) WHERE amount IS NOT NULL AND amount >= 0	Provides total revenue and financial transaction context.
Booking Status Mix	Percentage distribution of CONFIRMED, CANCELLED, PENDING	Tracks reservation conversion and cancellation rates.

13. Power BI Reporting Design

The Power BI analytical layer is strictly structured into a 3-page interactive report. Airline analysis is embedded organically across all pages using slicers and breakdown visual elements.

Page 1: Executive Overview

- **Purpose:** Provides high-level executive visibility into airline booking operations and revenue context.
- **KPI Cards:** Total Bookings, Total Revenue/Payments, Total Active Flights, Overall Confirmation Rate.
- **Visualizations:** Bar chart of Bookings by Airline; Donut chart of Booking Status Mix; Monthly Booking Volume Trend line chart.
- **Global Slicers:** Airline, Booking Status, Date Range.

Page 2: Flight Duration & Anomalies

- **Purpose:** Focuses on flight operational timings, overnight flight trends, and data quality duration anomalies.
- **KPI Cards:** Average Flight Duration (mins), Total Duration Anomalies, Total Overnight Flights.
- **Visualizations:** Clustered Column chart comparing Average Duration by Airline; Stacked Bar chart of Overnight vs Same Day flights; Anomaly Details data table.
- **Global Slicers:** Airline, Flight Day Type (Overnight/Same Day), Duration Anomaly Flag.

Page 3: Route Performance

- **Purpose:** Analyzes route-level demand, traffic corridors, and operational exceptions.
- **KPI Cards:** Total Active Routes, Top Route by Volume, Average Route Duration.
- **Visualizations:** Horizontal Bar chart of Top 10 Routes by Booking Volume; Matrix Table detailing Source, Destination, Flight Count, Average Duration, and Anomaly Count.
- **Global Slicers:** Source Airport, Destination Airport, Airline.

14. Assumptions & Business Rules

To ensure complete transparency, the implementation adheres to the following explicit business rules and technical assumptions:

- **1. Fallback Airline Mapping:** The 2-character flight ID prefix is authoritative for deriving missing/UNKNOWN airline names (SJ = SpiceJet, AI = Air India, UK = Vistara, 6F = IndiGo).
- **2. Record Deduplication:** Exact row duplicates in the source Flights sheet represent duplicate data logging and are safely purged.
- **3. Missing Attribute Categorization:** Missing booking statuses and missing passenger last names are explicitly categorized as 'Unknown' rather than dropping rows.
- **4. Financial Data Integrity:** Non-numeric or missing payment amounts are coerced to null. Financial values are never fabricated or imputed.
- **5. Duration Anomaly Threshold:** A 5-minute absolute difference between calculated elapsed duration and source supplied duration constitutes an Anomaly.
- **6. Anomaly vs. Delay Distinction:** A 'Duration Anomaly' reflects a source data discrepancy, NOT a real-world flight delay.
- **7. Primary Analytical Grain:** The analytical model is established at the individual booking level.
- **8. Infrastructure Scope:** Local Python/Pandas execution satisfies all project criteria; cloud platform migration represents a future deployment option.

15. Error Handling & Logging

The pipeline implements defensive programming practices to isolate data errors and maintain operational visibility:

- **Type Coercion & Trapping:** Datetime and numeric conversions utilize `errors='coerce'` to safely trap malformed strings into NaN/null values for validation auditing.
- **Validation Execution Report:** Each validation step outputs an execution summary detailing passed records, error counts, and flagged anomalies.
- **Pipeline Stage Logging:** Console logging tracks stage execution timestamps, row count shifts, and memory usage across ingestion, cleaning, transformation, and export.
- **Zero Financial Imputation:** Corrupted financial or timestamp fields are logged and isolated rather than imputed with artificial values.

16. Scalability & Performance Considerations

While implemented in Pandas for local execution, the architecture incorporates performance optimizations and presents a clear cloud migration roadmap:

- **Vectorized Pandas Execution:** All transformations, string extractions, and datetime math use C-optimized vectorized operations rather than inefficient row-by-row iteration.
- **Memory Management:** Unnecessary temporary calculation columns are dropped prior to exporting the final analytical CSV/parquet extract.
- **Future Cloud Migration Roadmap:** For multi-terabyte production scale, the exact functional logic can be deployed on:
 - **PySpark / Databricks:** Distributed dataframe processing across cloud worker clusters.
 - **Azure Data Factory / Fabric:** Enterprise orchestration, automated pipeline triggers, and Synapse Analytics integration.

17. Project Deliverables

The complete project submission comprises the following structured artifacts:

Deliverable Artifact	File Name / Repository Path	Description & Content
Pipeline Jupyter Notebook	notebooks/Airlines_Data_Engineering_Pipeline.ipynb	Executable Python pipeline covering Ingestion to Export.
Processed BI Datasets	data/processed/analytical_bookings_bi.csv	Cleaned, transformed, and PII-protected analytical extract.
Power BI Dashboard	powerbi/ASG_Airlines_Analytics.pbix	3-page interactive Power BI dashboard report.
Technical Documentation	docs/ ASG_Airlines_Data_Engineering_Documentation.docx	Formatted Word report submission.
PDF Technical Document	docs/ ASG_Airlines_Data_Engineering_Documentation.pdf	Compiled PDF report submission.
Repository Readme	README.md	Project overview, setup guide, and execution instructions.

18. Conclusion

The ASG Airlines End-to-End Data Engineering Pipeline successfully transforms raw, unstructured operational data into a highly polished, secure, and aggregation-ready analytical layer. By resolving complex data quality challenges—including duplicate purging, airline derivation, cross-midnight duration calculation, and duration anomaly detection—the pipeline guarantees high data fidelity for executive reporting.

Furthermore, the integration of SHA-256 cryptographic hashing and attribute masking enforces enterprise-grade data privacy governance. The delivered 3-page Power BI dashboard empowers operational leaders with actionable insights into route demand, flight timings, and carrier performance, serving as a robust data engineering solution for ASG Airlines.

Appendix A. Source Profiling Snapshot

Empirical summary of raw source sheet statistics profiled immediately upon workbook ingestion:

Source Sheet Name	Raw Row Count	Identified Data Quality Observations
flights	1,020	15 exact duplicate rows; 41 missing airline values; 31 'UNKNOWN' airline string entries.
payments	1,000	48 missing amount values; source column stored as non-numeric text object.
bookings	1,000	45 missing booking status values.
passengers	1,039	10 missing last_name values.

Appendix B. Airline Prefix Evidence in Source

Empirical distribution of 2-character flight ID prefixes extracted across raw flight records, validating the airline mapping logic:

Extracted Flight ID Prefix	Derived Carrier Identity	Observed Record Volume
6F	IndiGo	273 records
AI	Air India	260 records
SJ	SpiceJet	251 records
UK	Vistara	236 records